

Low-Dimensional Generative Models of Human Interception Movements

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Abstract

We study whether a compact representation of a participant's interception trials can generate the distribution of that participant's remaining movements. Five model families are evaluated on 4,732 trials from 28 participants using four participant-held-out folds. Spline+PCA reconstructs trajectories more accurately than a conditional variational autoencoder (CVAE), whereas CVAE has lower original-head timing errors and several lower distribution distances. A same-decoder control shows that the correct participant fingerprint improves generation over population and other-participant fingerprints at both three and eight latent dimensions. However, direct context measurements outperform latent probes on all 14 tested behavioural summaries under mean absolute error. Common timing heads reduce the CVAE initiation-time advantage, and minimum-jerk decomposition remains a secondary, assumption-sensitive evaluation. These results support useful participant information in the learned generator, while falling short of a validated low-dimensional account of individual movement strategy.

1 Introduction

Interception requires reaching an interception region at an appropriate time while the target moves [1]. The modelling objective is to reproduce distributions of initiation time, movement time and detailed kinematics across people. Movement can begin while information is still being gathered; describing a trajectory does not by itself identify the underlying decision process. Individual motor signatures motivate compact representations [2], although that work concerns a different task.

We distinguish reconstruction of an observed trial, prediction of its physical timing from its observed shape, and generation of new trials after enrolment from a held-out participant's context recordings. Only the third directly addresses distribution generation. Two or three latent coordinates are the desired compact interface; eight coordinates provide a higher-capacity comparison.

2 Data and study cohort

Participants moved a finger toward a fixed square as a circle approached it horizontally, aiming to arrive while the circle was inside. Six target start locations combine three distance/speed categories with left/right sides; speed varies continuously within category. Finger position was recorded in 3D at 240 Hz using a Polhemus Liberty tracker; target trajectories were sampled at 60 Hz. Following Prof. Friedman's instructions, only condition 2 (free eye movements) was analysed.

Of 4,763 available condition-2 recordings, four missing-arrival timeouts and 27 trials arriving more than 1 s after the target window were excluded, leaving 4,732 trials from 28 participants. The original project slides mention 29 participants; this report describes the 28 available in the analysed files. The retained outcomes are 3,006 Success, 326 Too late, 48 Too early and 1,352 fixation warnings. Outcome labels are not encoder inputs; early outcomes and warnings remain included provisionally.

3 Methods

3.1 Temporal and spatial representation

Frame counters establish time; duplicate positions are averaged and missing counters interpolated. Marker 5 identifies target appearance. MAT target trajectories identify motion onset, 0.20-0.50 s later (median 0.35 s), which anchors the model window. Positions are fourth-order, zero-phase Butterworth filtered at 10 Hz, translated to the initial window position and resampled to 100 phase points. The final model uses lateral x and forward y; the original 3D recordings remain available.

The model window runs from target motion to the final recorded finger sample, retaining post-motion waiting. The endpoint is an arrival proxy, on average 26.2 ms before MAT pressedTime (SD 3.6 ms). Finger onset is the first post-motion 3D speed crossing above 5 tracker units/s sustained for three frames. Initiation and movement durations are derived from these indices. Physical durations are withheld from the encoder and predicted separately; relative waiting remains visible in the phase-normalized shape. Units and event alignment require domain confirmation.

3.2 Models and training

The CVAE [3,4] encodes a 100 x 2 trajectory and five condition values (three start-category indicators, side and exact target speed) through two 256-unit ReLU layers into a diagonal Gaussian posterior. The decoder maps a latent draw and conditions to shape and two log-duration outputs. Its loss sums standardized coordinate squared errors, 20 times standardized log-timing squared errors and a KL term whose weight increases from 0 to 1 over 50 epochs. This is a weighted reconstruction objective, not a likelihood for the reported behavioural feature distributions.

Table 1. Model families. Matched latent size does not equalize parameter counts, objectives or timing-head capacity.

Family	Dimensions	Role in the comparison
CVAE	2, 3, 4, 8	Conditional variational shape/timing model
Spline+PCA	2, 3, 4, 8	Cubic splines, five interior knots; train-fitted PCA; Ridge timing head
Conditional AE	3, 8	Conditional network without stochastic bottleneck or KL
Unconditional VAE	3, 8	Variational network with task-condition input suppressed
Condition Ridge	No fingerprint	Condition-only shape/timing regression plus training residual sampling

Training uses participant-balanced sampling, Adam (initial learning rate 0.001; batch 64), gradient clipping at 5, validation learning-rate reduction and early stopping (patience 25), with a 150-epoch limit. Five of the 96 neural runs reached this limit. Scaling statistics are fitted on training data. The classical and neural settings are detailed in Appendix S1.

3.3 Participant-held-out evaluation

Each of four fixed folds uses 17 training, four validation and seven test participants; every participant is tested once. Seeds 42, 43 and 44 vary neural training randomness. The matrix has 48 CVAE, 24 conditional-AE, 24 unconditional-VAE, 16 spline and 12 condition-Ridge evaluations. These are all declared folds and seeds, not every possible partition. Test participants contribute disjoint, approximately half-session context/query sets stratified by start category and side. Their query trajectories never determine the fingerprint.

4 Reconstruction, timing and distribution fidelity

Reconstruction uses posterior means; trajectory MSE averages coordinate errors within trials and participants. Timing MAE averages absolute errors within participants, then participants equally. Generation draws 120 trials per participant around the context-mean latent code using a shared training-derived covariance. Query condition metadata determine the sampled task mixture; query shapes and timing labels are withheld. This evaluates enrolled participants under a known task mixture, not zero-shot prediction.

Mean KS averages two-sample KS statistics over 11 features and participants: initiation/movement time, peak speed, relative peak time, path length, straight-line distance, curvature, lateral deviation, heuristic speed-peak count and x/y endpoints. It is a discrepancy, not accuracy or a combined p-value. MMD² [8] and energy [9] compare standardized multivariate features using one training-only reference per fold; Appendix S1 gives the estimators.

Table 2. Out-of-fold, participant-balanced means. Neural results average three seeds. Lower error/distance is better. MSE is in tracker units squared; timing in ms. Enrolment is closed-set classification among seven enrolled people (chance 14.3%).

Model	n	MSE	Init. MAE (ms)	Move MAE (ms)	Mean KS	MMD ²	Enroll. (%)
Condition Ridge	-	1.234	98.5	72.7	0.277	0.126	-
Conditional AE	3	0.170	37.8	44.1	0.288	0.141	39.0
Conditional AE	8	0.026	39.9	42.7	0.259	0.111	46.5
CVAE	3	0.177	37.8	43.4	0.278	0.131	40.9
CVAE	8	0.032	34.8	44.2	0.219	0.067	50.7
Spline+PCA	3	0.097	56.2	51.2	0.325	0.181	36.4
Spline+PCA	8	0.023	55.6	50.2	0.255	0.115	48.6
Unconditional VAE	3	0.148	37.9	42.9	0.285	0.130	40.0
Unconditional VAE	8	0.033	36.0	42.2	0.215	0.064	50.8

Spline has lower reconstruction MSE than CVAE at n=3 (0.097 versus 0.177) and n=8 (0.023 versus 0.032). CVAE has lower original-head timing MAE, mean KS and MMD² at these dimensions. The paired CVAE-spline contrasts shown in Appendix S2 survive BH correction across the 56 main tests. They establish an endpoint-dependent tradeoff, not a uniformly superior model.

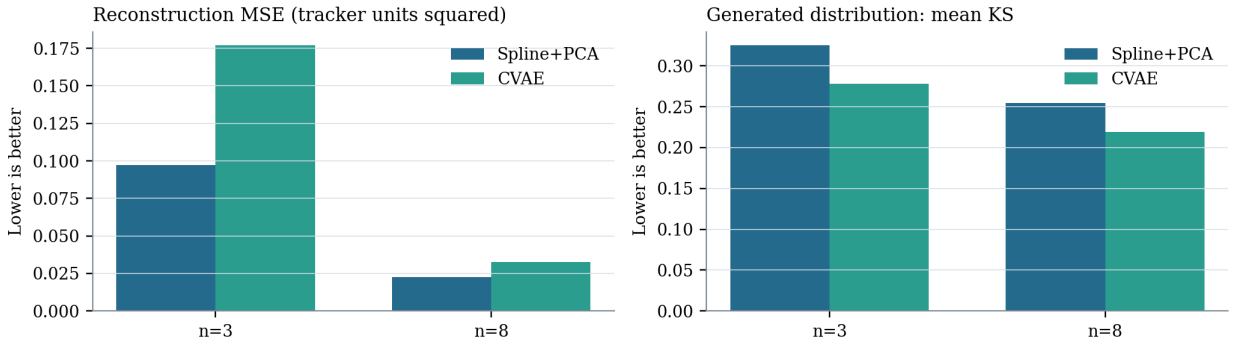


Figure 1. Selected benchmark endpoints; all panels show errors or discrepancies. The complete dimension sweep is in Appendix S2.

At n=3, CVAE does not improve mean KS, MMD or energy over condition Ridge after correction; at n=8 it improves all three. Conditional AE at n=8 reconstructs better than CVAE (MSE 0.026 versus 0.032), while CVAE improves KS (0.219 versus 0.259) and MMD² (0.067 versus 0.111). Unconditional VAE remains competitive, so neither conditioning nor the variational objective is uniformly beneficial.

5 Does the correct person's fingerprint help?

A post-review control isolates the fingerprint supplied to the same trained CVAE. For each of the 24 $n=3/8$ fold/seed checkpoints, we keep the decoder, 120 condition draws, latent-noise draws, shared covariance and distance reference fixed. We substitute (i) the participant's own context centroid, (ii) the equal-person average of the 17 training-context centroids, or (iii) each of the other six held-out context centroids. The wrong-person score averages six separate distances; it does not pool their generated distributions.

Table 3. Same-decoder fingerprint controls. Values average seeds within people, then all 28 people. Own-fingerprint generation reproduces the existing benchmark before controls are compared.

n	Fingerprint supplied	Mean KS	Energy	MMD ²
3	Correct participant	0.278	0.745	0.131
3	Training-participant average	0.311	1.016	0.175
3	Other participants (mean distance)	0.338	1.273	0.220
8	Correct participant	0.219	0.430	0.067
8	Training-participant average	0.290	0.919	0.149
8	Other participants (mean distance)	0.317	1.196	0.199

The correct fingerprint improves all three distances against both controls at both dimensions (12 comparisons; all BH-adjusted $p < 0.0004$). At $n=3$, mean KS improves over the population centre for 22/28 participants and over other-person centres for 27/28; at $n=8$ the corresponding counts are 27/28 and 27/28. Mean KS falls from 0.311 to 0.278 against the population control at $n=3$, and from 0.290 to 0.219 at $n=8$.

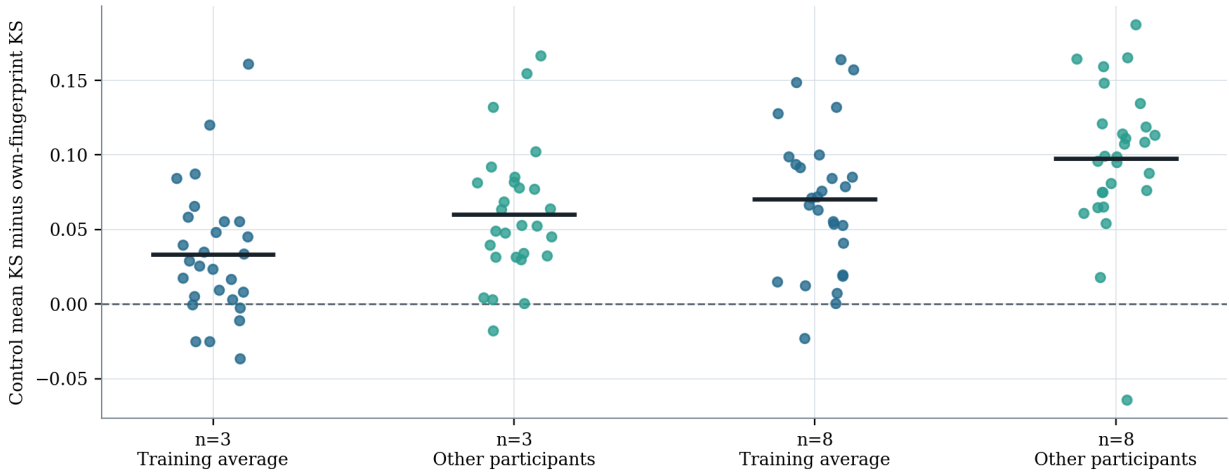


Figure 2. Participant-level change in mean KS when using the correct rather than a control fingerprint. Each point is one participant after averaging seeds; positive values favour the correct fingerprint. Lines mark zero and the participant mean.

This supports a specific claim: personal context contains information that improves this generator's within-session distribution predictions. It does not establish that the fingerprint recovers a cognitive strategy, stays stable across sessions, or reproduces every behavioural distribution. A useful fingerprint control can coexist with incomplete absolute fidelity and with competitive unconditional models.

5.1 Statistical scope

Paired two-sided Wilcoxon tests use one seed-averaged value per participant. The 56 main and 12 fingerprint tests use Pratt zero handling, SciPy 1.16.3 method='auto', and differences rounded to 12 decimal places before ranking [12]; effect-size means remain unrounded. BH correction [10] is applied separately to declared comparison families. Overlapping training folds mean participant errors are not fully independent [11]; these post-review controls support a within-cohort comparison, not an independent population replication.

6 Timing sensitivity and behavioural prediction

6.1 Comparing timing heads

The original CVAE timing head is nonlinear; spline uses linear Ridge. A sensitivity fits the same two-layer 64-unit MLP to frozen CVAE or spline codes plus conditions, using training/validation data only. A second sensitivity fits a positive multiplicative timing calibration to validation MAE. These comprise 48 additional small MLP fits and calibration checks, not retraining of the generative models.

Table 4. Timing sensitivities; BH correction covers eight paired comparisons. Calibration changes these sensitivity predictions only.

Head comparison	n	Endpoint	CVAE ms	Spline ms	Adjusted p
Common MLP	3	initiation	41.4	45.8	0.0076
Common MLP	3	movement	43.9	46.1	0.1872
Common MLP	8	initiation	42.7	47.9	0.0423
Common MLP	8	movement	44.8	44.8	0.8314
Validation calibration	3	initiation	37.2	44.4	<0.0001
Validation calibration	3	movement	44.3	50.9	<0.0001
Validation calibration	8	initiation	34.5	53.6	<0.0001
Validation calibration	8	movement	44.7	50.5	0.0001

With a common MLP, the CVAE initiation-time advantage is smaller but remains detectable; movement-time differences do not survive correction. CVAE codes were learned with timing supervision and spline codes were not, so the common head does not isolate representation geometry from the training objective. Validation calibration likewise cannot assign a percentage of the advantage to a particular cause.

6.2 Compact probes versus direct context measurements

Validation-tuned Ridge probes predict means and SDs of seven query behaviours from context fingerprints, refitting on 21 development participants before testing. For CVAE $n=3$, initiation-mean pooled R^2 is 0.723 and peak-speed-mean R^2 is 0.356; 10/14 pooled and 12/14 mean-fold scores are negative. Pooled and mean-fold R^2 have different denominators and are reported separately in Appendix S4.

Table 5. Summary prediction controls on the same query targets. The constant uses the 21 development participants; direct context copies the corresponding measured context mean or SD. Values average seed-specific MAEs.

Query-summary target	CVAE $n=3$ MAE	CVAE $n=8$ MAE	Direct context MAE	Development constant MAE
Mean initiation (ms)	14.78	14.71	9.27	30.38
Mean peak speed (tracker units/s)	7.74	8.76	1.73	11.88

Direct context has lower average MAE on all 14 targets at both dimensions. Its pooled R^2 is 0.912 for mean initiation and 0.976 for mean peak speed. This is a descriptive comparison, not a new significance claim. Direct measurement retains the target-specific summaries (14 values jointly), whereas the latent probe shares three or eight coordinates across targets. Thus personal information helps generation, but the tested compression does not improve these summary predictions over measuring the available context directly.

7 Minimum-jerk component fidelity

As a secondary description, we approximate movement velocity using one to four overlapping minimum-jerk pulses [5-7]. Each pulse has an onset, duration and planar displacement. Counts are selected by normalized fit error (smallest k at error at most 0.05, then below 0.10, otherwise minimum error); BIC is an alternative. The implementation adapts Friedman's code and is not identical to the published scattershot procedure. Components and heuristic speed peaks are different quantities; neither directly counts decisions.

The original recorded/generated fits had unequal optimization budgets and input processing. The post-review rerun applies the same downstream operator to the recorded model-target window and generated window: 100-point go-to-end input, timing-based extraction of 100 movement points, restoration of physical sample rate, low-pass filtering and two optimizer restarts with at most 400 function evaluations. All 2,376 query recordings and the same 1,680 generated samples (10 per person/seed/dimension) are fitted. Upstream acquisition filtering is retained in the recorded target; this is a comparison of the model representation, not unprocessed biological components.

Table 6. Matched-procedure component discrepancies. The empirical sampling reference uses 500 replicates at each participant's query size and ten generated samples, averaging three seed-like draws. It is a plug-in reference, not a formal goodness-of-fit p-value or hard floor.

Component discrepancy	n=3	n=8	Sampling reference
Fit error KS	0.339	0.308	0.266
First duration KS	0.455	0.409	0.267
First displacement amplitude KS	0.377	0.351	0.267
Secondary amplitude fraction KS	0.287	0.286	0.197
Overlap KS	0.267	0.268	0.195
Count total variation	0.214	0.232	0.146
Count JSD	0.076	0.087	0.047

Count total variation falls from historical 0.353/0.355 to 0.214/0.232 at $n=3/8$, weakening the earlier mismatch magnitude. All seven discrepancies remain above the empirical reference's 95th percentile, which excludes fitting and training uncertainty. All 4,056 selected fits converged; 505 fits had a nonconverged alternative order (Appendix S5). These are evaluation changes, not new neural training.

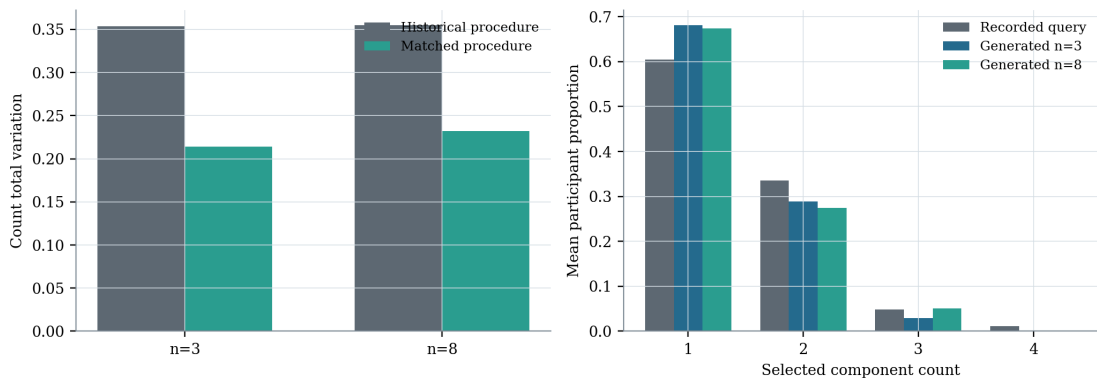


Figure 3. Legacy versus matched-procedure count discrepancies and selected-count proportions. The matched results are separate from the main 11-feature benchmark.

A fixed sensitivity subset (56 recorded and 112 generated trajectories) uses four restarts and 700 evaluations; selected-count agreement with the matched base is 100.0% and 99.1%, respectively. Of nine paired $n=3$ versus $n=8$ component endpoints, 0 survive BH correction. Historical 167 ms-bound and 5 Hz sensitivities also changed counts materially (Appendix S5). The component results remain conditional on fitting assumptions and finite sampling.

8 Discussion and limitations

The study supports an endpoint-dependent conclusion. Spline remains the stronger shape compressor; CVAE improves several distribution endpoints over spline and, at $n=8$, condition Ridge. Correct-person fingerprint controls establish useful participant information in this generator. Direct context summaries nevertheless outperform the tested latent probes, so these results do not establish a sufficient two- or three-coordinate account of a person's behavioural distributions.

8.1 Conditioning, clustering and interpretation

The in-sample trajectory K-means analysis gives $ARI=0.052$ and permutation $p=0.00498$ (200 permutations), indicating weak non-random structure rather than clean participant clusters. Closed-set enrolment reaches 40.9% at $n=3$ and 50.7% at $n=8$ against seven enrolled candidates; this is distinct from generation and cannot establish cross-session stability. The condition-stratum trajectory diagnostic gives Holm-adjusted $p=1.000$ and 0.132 at $n=3/8$, and none of 44 feature-level comparisons survives BH correction. Therefore the condition sliders remain exploratory. Latent-feature heatmaps show within-model associations, not independently validated strategy axes (Appendix S6).

8.2 Event definitions and retained early movement

There are 357 zero-initiation labels (7.5%): the post-target-motion speed threshold is immediately satisfied, with no missing-onset fallback among these trials. Of these, 109 show at least 50 ms of pre-motion threshold exceedance in filtered planar speed, and 100 in raw speed. Movement before target motion is distinct from starting while the moving target can supply information. These observations are retained and do not determine their cause; the encoder still begins at target motion. Event synchronization, endpoint validity and intended movement permissions need Prof. Friedman's confirmation.

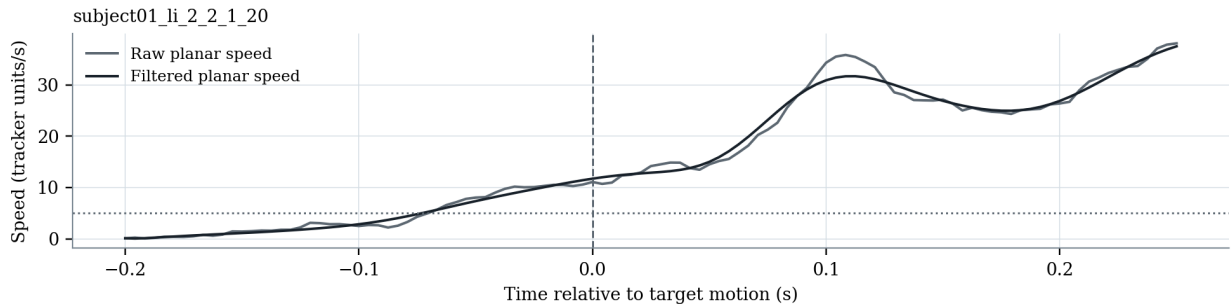


Figure 4. An audited example with raw and filtered planar speed around target-motion time. Threshold exceedance is descriptive evidence, not a diagnosis of noise or anticipation. Full examples and selection rules are in Appendix S7.

Rare timing failures also remain visible: CVAE initiation predictions reach about 11.5 s, without post-hoc clipping to improve the benchmark. The 74.6% statistic associated with zero-onset trials is a share of centred log-target variance, not observed trained-model loss or a demonstrated cause of outliers. The condition-2 fixation warning, the 1 s late exclusion, tracker units, onset threshold and movement-window choice remain acquisition/analysis assumptions, not established behavioural facts.

All participants come from one dataset; folds share training people. No learning-curve experiment establishes that participant count caused model failure, and no independent cohort validates the learned representation. Frozen-model controls and matched refits are explicitly post-review analyses. Stronger claims about cognition, stable identity or extrapolation require further targeted evidence rather than more favourable examples.

9 Conclusion

A participant's context-derived fingerprint improves generation with the tested CVAE, while reconstruction, summary prediction and component fidelity expose complementary limitations. The results justify a compact generative interface as an experimental model and support further evaluation of movement distributions. They do not yet justify a complete or uniquely interpretable model of individual interception strategy.

Reproducibility and acknowledgments

The evaluation covers all 124 declared model runs and preserves the original checkpoints, predictions and component fits. The matched component refits and fingerprint controls reuse these checkpoints; they do not retrain the neural models or change exclusions. Fold assignments, seeds, settings, checkpoint hashes and participant-level result tables accompany the code and appendix. Source histories record dirty-worktree states for two model families, so reproducibility is tied to saved artifacts and hashes rather than an unsupported clean-commit claim.

We thank Moni Shahar for project guidance and Prof. Jason Friedman for the experiment, data and methodological resources. AI tools assisted software development and report preparation; the authors remain responsible for understanding, verification and interpretation. A separate appendix contains the full dimension sweep, paired tests, all behavioural probe targets, convergence diagnostics, sensitivity checks, representative trajectories and execution instructions.

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